Dam Hong Phuc

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Summary

AI Solution Architect with **3+ years** of experience who goes beyond code: **engages customers** to find the real problem, turns **insight** into a focused product, then **measures and improves** it every release. Sole engineer and customer-facing owner of a Fortune 500 HR agent, from first quality baseline to zero-downtime migration; founding tech lead of an AI EdTech product.

Startup Product

Wizzdom **July 2025 – December 2025**

Technical Leader & Founding Member

- **Pivoted on evidence**: shipped a student AI-tutor MVP in **6 hours**; low adoption led us to pivot to paying teachers.
- **Customer discovery**: interviewed **20–30 tutoring teachers** in person and online; grading took 2–3 hours per class or paid assistants. They wanted cheap, fast grading to teach more students, not analytics, so we cut the roadmap to that job.
- **Sole engineer, full stack**: **Gemini** (vision + reasoning) splits each exam and writes a rubric per question; **Mathpix** OCRs handwritten math, Gemini finds each answer, a fuzzy match ties it to its question, and Gemini grades step by step, accepting alternative solutions. Stack: **React** + TypeScript on Vercel, **Python** on Lambda behind API Gateway, **AWS** (S3 → EventBridge → SQS → Lambda, RDS PostgreSQL, SES), Google OAuth.
- **Measured, human in the loop**: benchmarked against teacher grading (v1: **3x faster**, ∼70% agreement); teachers review, edit or re-grade, with AI notes on the handwriting, then approve. **1,000+ papers** graded; paying, returning teachers.

Professional Experience

FPT Software **January 2026 – Present**

AI Solution Architect

- **Project: Enterprise HR Conversational AI Platform – Covestro AG**
  - **Solution Architect**, not only delivering but also owning the customer relationship: worked directly with the Fortune 500 client's product owner and HR users on root-cause analysis and trade-off decisions.
  - **Agent platform**: re-architected a monolith into **microservices**: a stateless **agent-as-config LangGraph** engine on Lambda containers (ReAct loop, Guardrail in/out, versioned prompts and tool allowlists) calling **MCP** services for ServiceNow, Workday and knowledge; every tool call audited in DynamoDB.
  - **Retrieval & ingestion**: **Bedrock Knowledge Bases** + OpenSearch Serverless, Cohere embeddings, hybrid search, reranking, per-turn metadata filtering, permission-scoped **semantic cache** on ElastiCache; **event-driven ingestion** (EventBridge → SQS + DLQ → Lambda) with Claude vision and hierarchical, table-aware chunking.
  - **Evaluation-driven delivery**: RAGAS baseline, recall@k harness and **LLM-as-judge** parity gates reviewed by HR; **LLM regression suite** on every prompt, model or retrieval change; Langfuse token, cost and latency tracing.
  - **Zero-downtime migration**: Azure OpenAI → **Bedrock Claude**, validated on a cross-account PROD-mirror KB; SSM switch cutover with rehearsed rollback; stateful moves via CloudFormation stack refactor; **CI/CD** on GitHub Actions (OIDC, `cdk diff` gate, CodeQL, smoke tests, feature flags).
- **Project: Enterprise AI Contract Intelligence System**
  - Rebuilt a mega-prompt contract parser as a **skill-based framework**: Python filters context first, prompt chaining with **OpenSearch** RAG only when unsure: **+35% extraction accuracy**, ∼**5**× **fewer tokens**. Model routing on **AWS Bedrock** with SSE streaming; async SQS/Lambda batches where one failed contract never stops the rest.

HTI Group **April 2025 – December 2025**

AI Engineer & LLMOps

- **Project: Enterprise Agentic Chatbot & LLMOps Infrastructure**
  - **Multi-agent system**: a supervisor on **LangGraph** routes each query to Knowledge, Action or ChitChat agents, each running its own **ReAct loop** with session memory and parallel async tool calls.
  - **Agentic retrieval & actions**: Knowledge agent calls vector, full-text, graph and tabular tools in parallel until context is sufficient; Action agent checks parameters and **RBAC** before calling external APIs (Discord, IT tickets).
  - **Ingestion**: Docling layout parsing of PDF, Word and Excel; **hierarchical semantic chunking** with LLM context injection for leaf chunks; LLM-extracted **knowledge-graph** triplets; BGE-M3 embeddings.
  - **POC to AWS production**: FastAPI and LangGraph services on **ECS Fargate** behind API Gateway with gRPC streaming; async tools on Lambda; Bedrock via VPC endpoints; S3-triggered ingestion; CI/CD with **AWS CDK**.

HG Group **May 2023 – March 2025**

AI Engineer

- **HG Chatbot**: enterprise assistant with multi-format parsing, **hybrid retrieval** and **cross-encoder reranking**.
- **HG Render**: multimodal recommendation engine (**Qdrant** + GPT) assembling videos; async jobs on **Celery**, **Redis**.
- **HIG**: fine-tuned **FLUX.1** with **LoRA** for brand design styles; curated image datasets for training.

Technical Skills

Backend & Frontend: Python (FastAPI, asyncio), React, TypeScript, REST, SQS, Celery, PostgreSQL, DynamoDB, Redis
Agents & Evaluation: LangGraph, MCP, LlamaIndex, context engineering, RAGAS, LLM-as-judge, A/B release gates, Langfuse
Cloud & Delivery: AWS (Bedrock, Lambda, ECS, RDS, S3, IAM/KMS), CDK, GitHub Actions, Docker, Claude Code
Certifications: AWS GenAI Developer – Professional · AWS Solutions Architect – Associate · Deep Learning Specialization